Specifications for the Description of Maize Germplasm Resources

Author-prepared English reference translation

This reference translation follows the Chinese crop descriptor specification used for the CropMaestro study. It is not an official English edition issued by the source platform. Original section numbers and category-code mappings are retained.

1 Scope

This specification establishes the descriptors for maize germplasm resources and their classification criteria.

It applies to the collection, documentation and conservation of maize germplasm resources; the development of data standards and data quality control specifications; and the establishment of databases and information-sharing network systems.

2 Normative references

The provisions of the following documents become provisions of this specification through reference herein. For dated references, subsequent amendments (excluding corrections) or revised editions do not apply to this specification. However, parties to agreements based on this specification are encouraged to investigate whether the latest editions of these documents can be used. For undated references, the latest editions apply.

ISO 3166 Codes for the Representation of Names of Countries

GB/T 2659 Codes for the Representation of Names of Countries and Regions

GB/T 2260 Codes for the Administrative Divisions of the People's Republic of China

GB/T 12404 Codes for the Administrative Affiliation of Organizations

GB/T 3543.1-7-1995 Rules for Agricultural Seed Testing

GB 1353-1999 Maize

GB/T 8613-1999 Maize for the Starch Fermentation Industry

NY/T 520-2002 Quality Protein Maize

NY/T 521-2002 High-Oil Maize

NY/T 597-2002 High-Starch Maize

NY/T 523-2002 Sweet Maize

NY/T 524-2002 Waxy Maize

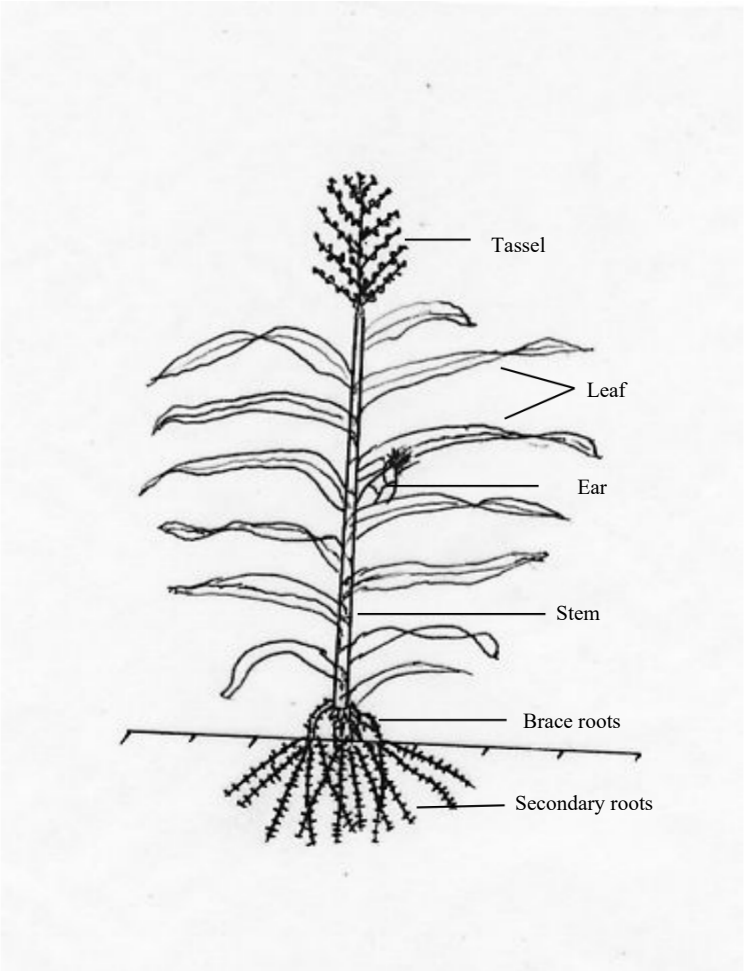
NY/T 522-2002 Popcorn

3 Terms and definitions

3.1 Maize

An annual herbaceous species in the family Gramineae, tribe Maydeae and genus Zea L., with the scientific name Zea mays L. Its Chinese common names include baogu, baomi, bangzi, yujiao and zhenzhumi. The chromosome number is 2n=2x=20. Maize plants are relatively tall and comprise roots, stems, leaves, tassels, ears and other parts (Figure 1).

Figure 1. Maize plant



3.2 Maize germplasm resources

Wild maize resources, landraces, composite varieties/synthetic populations, inbred lines, genetic stocks and other materials.

3.3 Basic information

Basic descriptive information on maize germplasm resources, including the national unified identifier, germplasm name, scientific name, place of origin and germplasm type.

3.4 Morphological and biological characteristics

Characteristics of maize germplasm resources such as phenological stages, botanical morphology and yield traits.

3.5 Quality characteristics

Nutritional quality traits of maize germplasm resources, including protein content, fat content, starch content and lysine content.

3.6 Abiotic stress tolerance

The ability of maize germplasm resources to adapt to or withstand abiotic stresses, including drought, cold, waterlogging, high planting density and salinity.

3.7 Disease and pest resistance

The ability of maize germplasm resources to adapt to or resist biotic stresses, including northern and southern corn leaf blight, Curvularia leaf spot, stalk rot, ear rot, sheath blight, maize dwarf mosaic and maize borers.

3.8 Combining ability

The concept of combining ability arose from research into the genetic basis of heterosis and the development of breeding methods for maize hybrids. Combining ability refers to the productivity or the magnitude of another trait of the first-generation hybrid produced by crossing one parental line with another. It is not expressed through the performance of the parent itself, but is estimated from the mean phenotypic value of the target trait, such as yield, in hybrids formed using that parent. General combining ability and specific combining ability are distinguished.

When an inbred line or variety is crossed with several other inbred lines or varieties, the mean yield or value of another trait expressed by the first-generation hybrids is termed its general combining ability (GCA). This reflects the general relationship between that parent and many other parents. Its magnitude is determined by the number of favourable gene loci in the parental line. It is controlled by additive gene effects and constitutes the heritable component. Parents with high GCA are more likely to produce high-yielding hybrids; GCA has therefore become an important criterion for selecting parents in maize breeding.

When a particular inbred line or variety is crossed with another particular inbred line or variety, the yield or value of another trait expressed by the first-generation hybrid is termed the specific combining ability (SCA) of that particular combination. This reflects the specific relationship between the two parents. It is determined by non-additive gene effects and is controlled by dominance, overdominance and epistasis. It is expressed only in particular hybrid combinations through interactions between alleles or between non-allelic genes from the two parents and constitutes the non-heritable component.

Combining ability evaluation in this specification refers to the evaluation of heritable general combining ability.

4 Basic information

4.1 National unified identifier

The unique identifier of the germplasm. For domestic maize germplasm, the national unified identifier consists of a 2-digit germplasm-type code, a 2-digit province or municipality code and a 4-digit serial number. For foreign germplasm, it consists of a 2-digit germplasm-type code, a 1-digit continent code, a 2-digit country code and a 3-digit serial number. Identifiers are assigned centrally by the lead institution responsible for cataloguing maize germplasm resources.

4.2 Genebank accession number

The number assigned to maize germplasm in the National Long-term Crop Germplasm Bank, consisting of "I1G" followed by a 5-digit serial number.

4.3 Introduction number

The number assigned to maize germplasm when introduced from abroad.

4.4 Collecting number

The number assigned to maize germplasm when collected in the field.

4.5 Germplasm name

The Chinese name of domestic germplasm or the Chinese translation of the name of germplasm introduced from abroad.

4.6 Germplasm name in a foreign language

The foreign-language name of germplasm introduced from abroad, or the name of domestic germplasm in Hanyu Pinyin.

4.7 Family

Gramineae.

4.8 Genus

Zea L.

4.9 Scientific name

The scientific name of maize is Zea mays L.

4.10 Country of origin

The name of the country, region or international organization from which the maize germplasm originated.

4.11 Province of origin

For domestic maize germplasm, the name of the province of origin; for introduced foreign germplasm, the name of the first-level administrative division in the country of origin.

4.12 Locality of origin

The names of the county, township and village of origin of domestic maize germplasm.

4.13 Elevation

The elevation of the place of origin of the maize germplasm. Unit: m.

4.14 Longitude

The longitude of the place of origin of the maize germplasm, in degrees (°) and minutes (′). The format is DDDFF, where DDD denotes degrees and FF denotes minutes.

4.15 Latitude

The latitude of the place of origin of the maize germplasm, in degrees (°) and minutes (′). The format is DDFF, where DD denotes degrees and FF denotes minutes.

4.16 Source location

For maize germplasm introduced from abroad, the name of the source country, region or international organization; for domestic germplasm, the names of the source province and county.

4.17 Original holding institution

The name of the institution that held the maize germplasm before it was submitted to the National Long-term Crop Germplasm Bank.

4.18 Accession number at the original holding institution

The germplasm number assigned by the institution that originally held the maize germplasm.

4.19 Pedigree

The genetic relationships of bred maize inbred lines and composite varieties/synthetic populations.

4.20 Breeding institution or breeder

The name of the institution or individual that developed the maize inbred line or composite variety/synthetic population.

4.21 Year of development

The year in which the maize inbred line or composite variety/synthetic population was successfully developed.

4.22 Breeding method

The method used to breed the maize inbred line or composite variety/synthetic population.

4.23 Germplasm type

Maize germplasm is classified into 6 types.

1	Wild germplasm
2	Landrace
3	Composite variety/synthetic population
4	Inbred line
5	Genetic stock
6	Other

4.24 Image

The filename of an image of the maize germplasm. The image format is .jpg.

4.25 Observation site

The name of the site where the morphological and biological characteristics of the maize germplasm are observed.

5 Morphological and biological characteristics

5.1 Sowing date

The date on which maize germplasm is sown for the evaluation of morphological and biological characteristics. Express the date as year, month and day in the format YYYYMMDD.

5.2 Emergence date

The date on which at least 50% of the shoots in the plot extend at least 3.0 cm above the soil surface. Express the date as year, month and day in the format YYYYMMDD.

5.3 Tassel emergence date

The date on which the tassel tip has emerged above the uppermost leaf in at least 50% of the plants in the plot. Express the date as year, month and day in the format YYYYMMDD.

5.4 Anthesis and pollen-shedding date

The date on which the tassels of at least 50% of the plants in the plot begin flowering and shedding pollen. Express the date as year, month and day in the format YYYYMMDD.

5.5 Silking date

The date on which silks emerge from the ear husks in at least 50% of the plants in the plot. Express the date as year, month and day in the format YYYYMMDD.

5.6 Maturity date

The date on which kernels on at least 90% of the plants in the plot pass from the late dough stage to full maturity: the milk line disappears from kernels in the middle of the ear; kernels lose water and harden; a black layer appears at their base; and the husks turn yellow. Express the date as year, month and day in the format YYYYMMDD.

5.7 Days from sowing to emergence

The number of days from the day after sowing to the emergence date. Unit: d.

5.8 Days from emergence to tassel emergence

The number of days from the emergence date to the tassel emergence date. Unit: d.

5.9 Days from tassel emergence to anthesis and pollen shedding

The number of days from the tassel emergence date to the anthesis and pollen-shedding date. Unit: d.

5.10 Days from anthesis and pollen shedding to silking

The number of days from the anthesis and pollen-shedding date to the silking date. Unit: d.

5.11 Days from silking to maturity

The number of days from the silking date to the maturity date. Unit: d.

5.12 Days to maturity

The number of days from the day after sowing to the maturity date. Unit: d.

5.13 Effective accumulated temperature (≥10°C)

The accumulated temperature at or above 10°C from the day after sowing to maturity. Unit: °C.

5.14 Planting density

The number of plants retained per hectare after final thinning. Unit: plants/hm^2.

5.15 Seedling leaf colour

The colour of the first maize leaf at the seedling stage.

1	Light green
2	Green
3	Dark green
4	Purple

5.16 Seedling coleoptile colour

Observe the colour of the coleoptile after the first leaf has unfolded.

1	Green
2	Green with purple streaks
3	Purple
4	Dark purple

5.17 Seedling vigour

The degree of plant vigour at the seedling stage.

1	Weak
2	Intermediate
3	Strong

5.18 Tillering capacity

The ability of the germplasm to produce tillers.

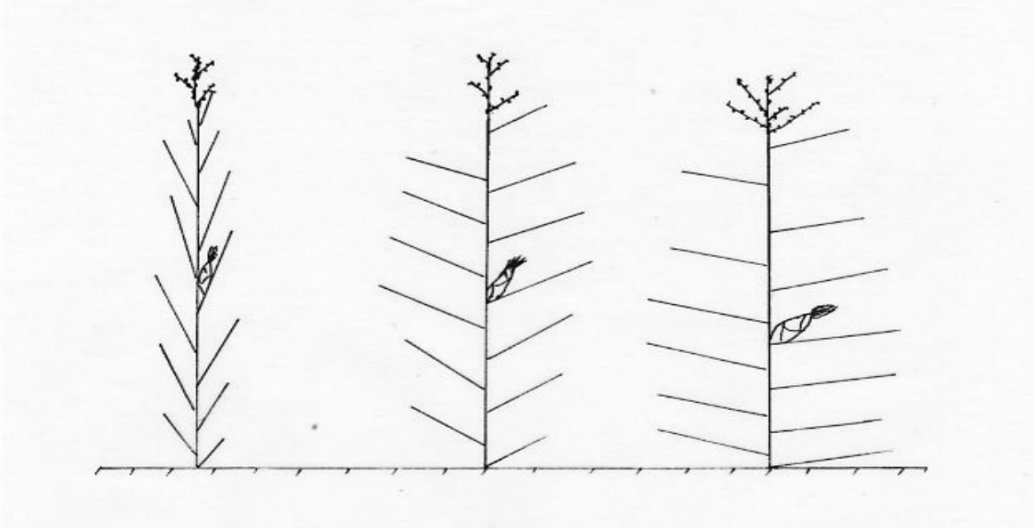
0	Absent
1	Weak
2	Intermediate
3	Strong

5.19 Plant architecture

Determined by the angle between the leaf blade and the main stem, also called the leaf insertion angle (Figure 2).

1	Compact
2	Intermediate
3	Spreading

Figure 2. Plant architecture



1 2 3

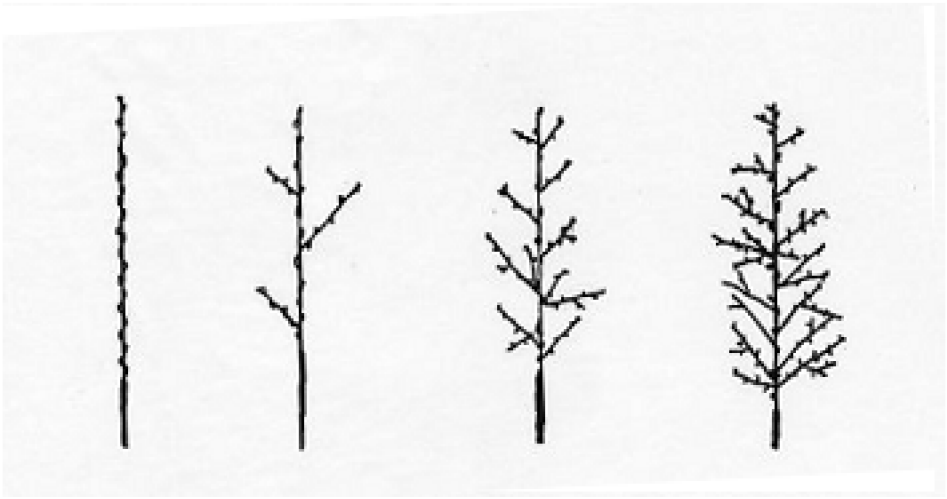
1 Compact; 2 Intermediate; 3 Spreading.

5.20 Number of primary tassel branches

The number of branches arising from the main axis of the male inflorescence (Figure 3).

0	Absent
1	Few
2	Intermediate
3	Many

Figure 3. Primary tassel branches



0 1 2 3

0 Absent; 1 Few; 2 Intermediate; 3 Many.

5.21 Tassel length

The length of the main axis of the male inflorescence. Unit: cm.

5.22 Tassel glume colour

The colour of the tassel glumes.

1	Green
2	Green with purple streaks
3	Purple

5.23 Anther colour

The colour of the anthers in the tassel.

1	Green
2	Light purple
3	Purple
4	Dark purple

5.24 Silk colour

The colour of the silks of the ear.

1	Yellow-green
2	Light red
3	Dark red
4	Variegated

5.25 Plant height

The height of the main stem from ground level to the top of the tassel. Unit: cm.

5.26 Ear height

The height from ground level to the node bearing the uppermost productive ear on the main stem. A productive ear is an ear that matures normally and produces at least 10 kernels; this definition also applies below. Unit: cm.

5.27 Number of leaves on the main stem

The total number of leaves borne on the main stem. Unit: leaves.

5.28 Leaf position of the uppermost productive ear

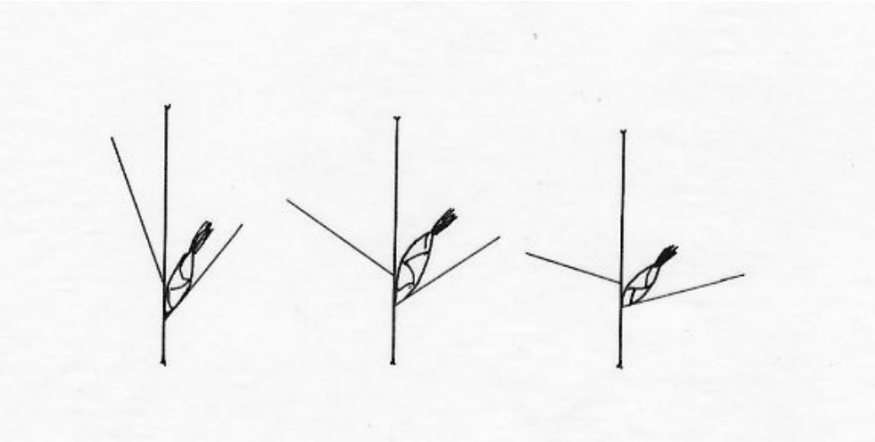
The leaf position, expressed as a leaf number, at which the uppermost productive ear is borne on the main stem.

5.29 Angle of the first leaf above the uppermost productive ear

The growth angle of the first leaf above the leaf subtending the uppermost productive ear on the main stem (Figure 4).

1	Erect
2	Intermediate
3	Horizontal

Figure 4. Angle of the first leaf above the uppermost productive ear



1 2 3

1 Erect; 2 Intermediate; 3 Horizontal.

5.30 Colour of the first leaf above the uppermost productive ear

The colour of the first leaf above the leaf subtending the uppermost productive ear on the main stem.

1	Pale green
2	Green
3	Dark green

5.31 Length of the first leaf above the uppermost productive ear

The length, from ligule to tip, of the first leaf above the leaf subtending the uppermost productive ear on the main stem. Unit: cm.

5.32 Width of the first leaf above the uppermost productive ear

The width at the midpoint of the first leaf above the leaf subtending the uppermost productive ear on the main stem. Unit: cm.

5.33 Brace-root development

The development of brace roots on the above-ground portion of the main stem.

0	Absent
1	Weak
2	Intermediate
3	Strong

5.34 Plant uniformity

The uniformity of growth and development among plants.

1	Uniform
2	Relatively uniform
3	Non-uniform

5.35 Number of productive tillers per plant

For tillering germplasm, the number of stems per plant that develop productive ears, excluding the main stem. Unit: tillers.

5.36 Number of productive ears per stem

The number of productive ears developed on a single stem. Unit: ears.

5.37 Double-ear percentage

The percentage of all plants that bear two or more productive ears on a single stem. Expressed as %.

5.38 Barren-plant percentage

The percentage of plants in the population with 0 productive ears, relative to the total number of plants. Expressed as %.

5.39 Stem diameter

The minor diameter at the midpoint of the third above-ground internode of a typical plant, excluding the leaf sheath. Unit: cm.

5.40 Root-lodging percentage

The percentage of plants in the population that are lodged from the roots at maturity (leaning angle >45º), relative to the total number of plants. Expressed as %.

5.41 Stalk-breakage percentage

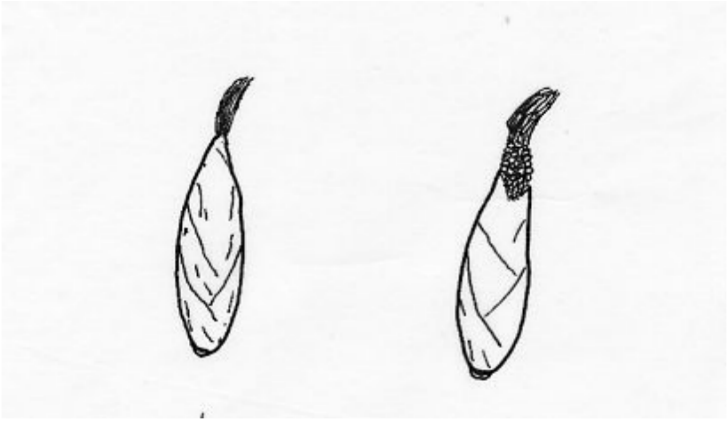
The percentage of plants in the population whose stalks have broken at maturity, relative to the total number of plants. Expressed as %.

5.42 Completeness of ear husk cover

The extent to which the ear is enclosed by its husks (Figure 5).

1	Complete cover
2	Incomplete cover

Figure 5. Ear husk cover



1 2

1 Complete cover; 2 Incomplete cover.

5.43 Ear-shank length

The length from the base of the ear to its point of attachment to the main stem. Unit: cm.

5.44 Ear-shank angle

The orientation of the ear shank, expressed as its angle to the main stem. Unit: º.

5.45 Fresh ear weight per plant

The total fresh weight of all productive ears on a plant. Unit: g.

5.46 Kernel moisture content at maturity

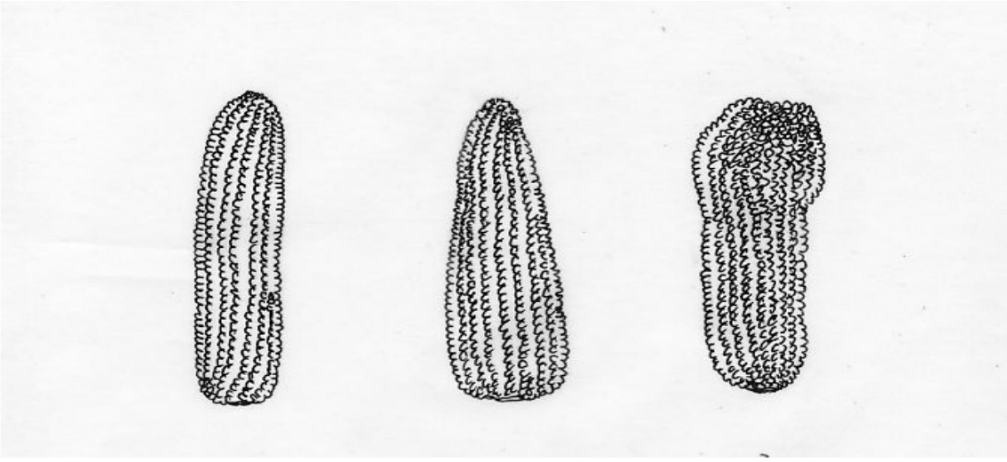
The moisture content of fresh kernels measured at harvest at maturity. Expressed as %.

5.47 Ear shape

The shape of the ear of the germplasm (Figure 6).

1	Cylindrical
2	Conical
3	Flattened

Figure 6. Ear shape



1 2 3

1 Cylindrical; 2 Conical; 3 Flattened.

5.48 Ear length

The length from the base to the tip of the ear. Unit: cm.

5.49 Barren-tip length

The length of the portion at the ear tip that bears no kernels. Unit: cm.

5.50 Ear diameter

The diameter at the midpoint of the ear; measure the minor diameter for flattened ears. Unit: cm.

5.51 Number of kernel rows per ear

The range of numbers of kernel rows per ear, expressed as even numbers.

5.52 Number of kernels per row

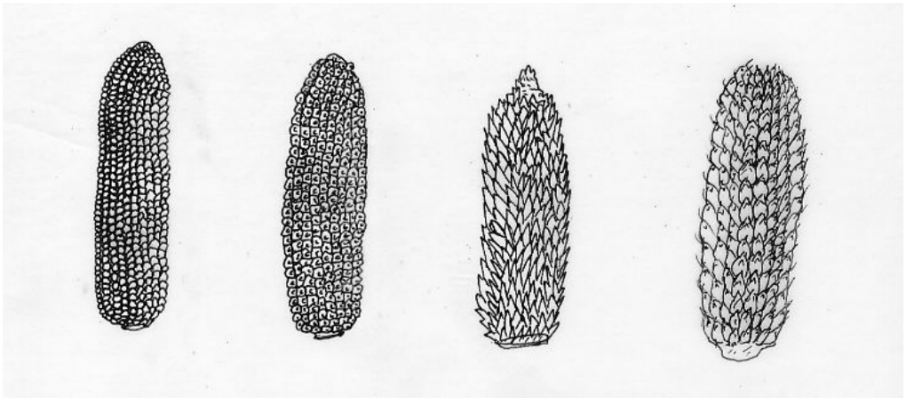
The number of kernels in each row on an ear. Unit: kernels.

5.53 Kernel type

The type of mature kernels (Figure 7).

1	Flint
2	Dent
3	Intermediate
4	Waxy
5	Sweet
6	Popcorn
7	Floury
8	Pod (tunicate)
9	Sweet-floury

Figure 7. Representative kernel types



1 2 6 8

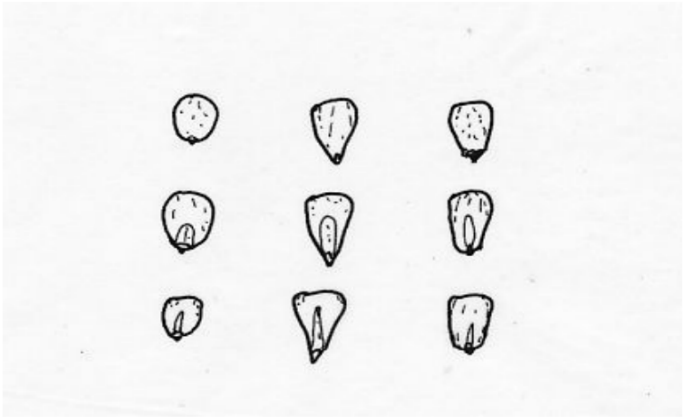
1 Flint; 2 Dent; 6 Popcorn; 8 Pod (tunicate).

5.54 Kernel shape

The shape of mature kernels (Figure 8).

1	Round
2	Wedge-shaped
3	Intermediate

Figure 8. Kernel shape



1 2 3

1 Round; 2 Wedge-shaped; 3 Intermediate.

5.55 Kernel size

The size of mature kernels.

1	Small
2	Medium
3	Large

5.56 Kernel colour

The external colour of mature kernels.

1	White
2	Light yellow
3	Yellow
4	Orange-yellow
5	Tangerine-yellow
6	Red
7	Blue
8	Purple
9	Black
10	White with red streaks
11	Yellow with red streaks
12	Yellow kernels with white streaks
13	Yellow kernels with red spots
14	White cap
15	Patterned
16	White/yellow
17	Yellow/white
18	Yellow/red
19	Red/yellow
20	Yellow/purple
21	Purple/yellow
22	Variegated

5.57 Cob colour

The colour of the cob of a mature ear.

1	White
2	Light red
3	Red
4	Purple
5	White/red
6	Red/white

5.58 Cob diameter

The diameter at the midpoint of the cob after maturity; measure the minor diameter for flattened cobs. Unit: cm.

5.59 Dry ear weight per plant

The total dry weight of all productive ears on a plant (naturally air-dried, with kernel moisture content taken as 13%; the same basis applies below). Unit: g.

5.60 Dry grain weight per plant

The total dry weight of all kernels produced by a single plant. Unit: g.

5.61 Shelling percentage

Kernel dry weight as a percentage of ear dry weight. Expressed as %.

5.62 Thousand-kernel weight

The weight of 1000 intact kernels determined by random sampling, with kernel moisture content taken as 13%. Unit: g.

5.63 Test weight

The weight of kernels occupying one litre, with kernel moisture content taken as 13%. Unit: g/L.

5.64 Grain yield

The grain yield per hectare calculated from the total kernel dry weight measured in the sampled plot. Unit: kg/hm^2.

5.65 Fresh biomass yield

The yield per hectare calculated from the total fresh weight of the above-ground parts of all plants in the plot, measured at maize harvest. Unit: kg/hm^2.

5.66 Dry biomass yield

The yield per hectare calculated from the total dry weight of the above-ground parts of all plants, determined by sampling the plot. Unit: kg/hm^2.

5.67 General combining ability

The mean yield or value of another trait expressed by the first-generation hybrids obtained by crossing a particular germplasm accession with many inbred lines or varieties. Classified into 3 categories.

1	Weak
2	Intermediate
3	Strong

6 Quality characteristics

6.1 Total starch content

The mass of starch in the kernels as a percentage of kernel mass. Expressed as %.

6.2 Amylose content

The mass of amylose in the kernels as a percentage of total starch mass. Expressed as %.

6.3 Amylopectin content

The mass of amylopectin in the kernels as a percentage of total starch mass. Expressed as %.

6.4 Soluble sugar content

The amount of soluble sugars in the kernels as a percentage of total kernel sugars. Expressed as %.

6.5 Crude protein content

The mass of protein in the kernels as a percentage of kernel mass. Expressed as %.

6.6 Lysine content

The mass of lysine in the kernels as a percentage of kernel protein mass. Expressed as %.

6.7 Crude fat content

The mass of fat in the kernels as a percentage of kernel mass. Expressed as %.

6.8 Fatty acid value (KOH)

The number of milligrams of potassium hydroxide required to neutralize the free fatty acids in 100 g of kernels and their products. Unit: 10^{-2} mg/g.

6.9 Vitamin E content

The mass of vitamin E in the kernels as a percentage of kernel mass. Expressed as %.

6.10 Popping percentage

The number of kernels that expand and pop into popcorn upon heating as a percentage of the total number of kernels. Expressed as %.

6.11 Popping expansion ratio

The ratio of the volume of popcorn after popping to the volume of the unpopped kernels. Expressed as a multiple.

6.12 Eating quality

A quality indicator for sweet and waxy maize intended for fresh consumption. It is evaluated by tasting and considers the sweetness, stickiness, pericarp thickness, flavour, mouthfeel and other attributes of the sample. Classified into 3 categories.

1	Excellent
2	Intermediate
3	Poor

7 Abiotic stress tolerance

7.1 Drought tolerance at germination

The ability of maize seeds to germinate under drought stress.

1	Very strong
3	Strong
5	Intermediate
7	Weak
9	Very weak

7.2 Drought tolerance at the seedling stage

The ability of maize seedlings to tolerate or resist drought stress.

1	Very strong
3	Strong
5	Intermediate
7	Weak
9	Very weak

7.3 Drought tolerance at anthesis and pollen shedding

The ability of maize to tolerate or resist drought stress during anthesis and pollen shedding.

1	Very strong
3	Strong
5	Intermediate
7	Weak
9	Very weak

7.4 Drought tolerance during grain filling

The ability of maize to tolerate or resist drought stress during grain filling.

1	Very strong
3	Strong
5	Intermediate
7	Weak
9	Very weak

7.5 Cold tolerance at germination

The ability of maize seeds to germinate under low-temperature stress.

1	Very strong
3	Strong
5	Intermediate
7	Weak
9	Very weak

7.6 Cold tolerance at the seedling stage

The ability of maize seedlings to tolerate or resist low temperatures.

1	Very strong
3	Strong
5	Intermediate
7	Weak
9	Very weak

7.7 Cold tolerance during grain filling

The ability of maize to tolerate or resist low temperatures during grain filling.

1	Very strong
3	Strong
5	Intermediate
7	Weak
9	Very weak

7.8 Salt tolerance

The ability of maize germplasm to tolerate or resist salt stress.

1	Very strong
3	Strong
5	Intermediate
7	Weak
9	Very weak

7.9 Waterlogging tolerance at the seedling stage

The ability of maize germplasm to tolerate or resist waterlogging at the seedling stage.

1	Very strong
3	Strong
5	Intermediate
7	Weak
9	Very weak

7.10 Tolerance to high planting density

The ability of maize germplasm to tolerate dense planting.

1	Very strong
3	Strong
5	Intermediate
7	Weak
9	Very weak

8 Disease and pest resistance

8.1 Northern corn leaf blight resistance

The degree of resistance of maize to the disease caused by Exserohilum turcicum (Pass.) Leonard et Suggs, with the sexual state Setosphaeria turcica (Luttrell) Leonard et Suggs, characterized primarily by large lesions on the leaves.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.2 Southern corn leaf blight resistance

The degree of resistance of maize to the disease caused by Bipolaris maydis (Nisikado et Miyake) Shoemaker, with the sexual state Cochliobolus heterostrophus (Drechsler) Drechsler, characterized primarily by small lesions on the leaves.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.3 Curvularia leaf spot resistance

The degree of resistance of maize to the disease caused by fungi of the genus Curvularia, principally Curvularia lunata (Walk.) Boed., with the sexual state Cochliobolus lunatus Nelson et Haasis, characterized primarily by small lesions on the leaves.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.4 Gray leaf spot resistance

The degree of resistance of maize to the disease caused by fungi of the genus Cercospora, including Cercospora zeae-maydis Tehon et Daniels, characterized primarily by leaf lesions.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.5 Northern corn leaf spot resistance

The degree of resistance of maize to the disease caused by Bipolaris zeicola (Stout) Shoemaker, with the sexual state Cochliobolus carbonum Nelson, characterized primarily by leaf lesions.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.6 Physoderma brown spot resistance

The degree of resistance of maize to the disease caused by Physoderma maydis (Miyabe) Miyabe, characterized primarily by brown spots on the stalks and leaves.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.7 Rust resistance

The degree of resistance of maize to common rust and southern rust, caused respectively by Puccinia sorghi Schwein. and Puccinia polysora Underw. and characterized primarily by rust pustules on the leaves and stalks.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.8 Head smut resistance

The degree of resistance of maize to the disease caused by Sphacelotheca reiliana (Kühn) Clinton, characterized primarily by smutted ears and tassels.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.9 Common smut resistance

The degree of resistance of maize to the disease caused by Ustilago zeae (Beckm.) Unger, characterized primarily by black galls on the ears, stalks and leaves.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.10 Sheath blight resistance

The degree of resistance of maize to the disease caused by fungi of the genus Rhizoctonia, including Rhizoctonia solani Kühn, with the sexual state Thanatephorus cucumeris (Frank) Donk, and Rhizoctonia zeae Voorhees, with the sexual state Waitea circinata Warcup et Talbot, characterized primarily by discolouration of leaf sheaths on the stalk.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.11 Stalk rot resistance

The degree of resistance of maize to diseases caused primarily by fungi such as Pythium spp. and Fusarium spp., characterized by decay at the stalk base that leads to whole-plant wilting, with the plant remaining green or turning yellow.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.12 Ear rot resistance

The degree of resistance of maize to diseases caused primarily by Fusarium fungi, including Fusarium verticillioides (Sacc.) Nirenberg = F. moniliforme Sheldon, with the sexual state Gibberella fujikuroi (Sawada) Wollenw., and by Aspergillus spp., characterized primarily by rotting kernels on the ear.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.13 Crazy top resistance

The degree of resistance of maize to the disease caused by Sclerophthora macrospora (Sacc.) Thirum., C.G. Shaw & Naras., characterized primarily by abnormal growth of the tassels and ears.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.14 Maize rough dwarf disease resistance

The degree of resistance of maize to the disease caused by Maize rough dwarf virus, characterized primarily by whole-plant dwarfing and thickened internodes.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.15 Maize dwarf mosaic resistance

The degree of resistance of maize to the disease caused by Sugarcane mosaic virus - Maize dwarf strain B, characterized primarily by whole-plant dwarfing and mosaic symptoms on the leaves.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

8.16 Maize borer resistance

The degree of resistance of maize to infestation by the Asian corn borer, Ostrinia furnacalis (Guenée), characterized primarily by feeding damage to the leaves.

1	Highly resistant (HR)
3	Resistant (R)
5	Moderately resistant (MR)
7	Susceptible (S)
9	Highly susceptible (HS)

9 Other characteristics

9.1 Fertility

The ability of maize to produce normal female and male gametes and produce normal offspring during sexual reproduction. Sterility is the genetic phenomenon in which normal ears, silks or female gametes cannot be produced; normal anthers, pollen or male gametes cannot be produced; or female and male gametes cannot unite normally to produce offspring. It may be classified as female sterility or male sterility. Male-sterile lines can be used to produce maize hybrids.

1	Fertile
2	Sterile

9.2 Type of male sterility

The type of genetic control of male sterility in male-sterile germplasm.

1	Nuclear male sterility
2	Cytoplasmic male sterility
3	Male sterility due to nuclear-cytoplasmic interaction

9.3 Cytoplasmic male-sterility group

The cytoplasmic group of male-sterile germplasm in which sterility is under cytoplasmic control.

1	T group
2	C group
3	S group

9.4 Karyotype

A formula representing chromosome number, size, morphology and structural characteristics.

9.5 DNA fingerprint

The DNA fingerprint profile of each maize germplasm accession, identified using molecular markers.

9.6 Use

The principal use of the maize germplasm.

1	Food
2	Feed
3	Industrial use

9.7 Notes

Detailed explanations of special descriptors or special codes for the maize germplasm that are not covered above.

Translator's notes on the source

5.13: The source specifies °C as the unit and does not provide a calculation formula for effective accumulated temperature.

5.56: The source lists these colour categories separately but does not define their boundaries or the order of components in slash-separated colour labels. 'Patterned' and 'variegated' retain separate source codes without adding a pattern definition.

8.3: The source spelling 'Nalson' has been standardized to 'Nelson', following GBIF name record 2616015.

8.4: The source spelling 'Cercospora zea-maydis' has been standardized to 'Cercospora zeae-maydis', following EPPO entry CERCZM.

8.10: The source spelling 'Weitea circinata' has been standardized to 'Waitea circinata', following GBIF name record 5249953.

8.12: The source spelling 'Fusarium verticilloides' has been standardized to 'Fusarium verticillioides', following GBIF name record 5251653.

8.13: The source spelling 'Sclerophotora macrospora (Saccado)' has been standardized to 'Sclerophthora macrospora (Sacc.)', following GBIF name record 3203570.

5.53: The source lists sweet-floury as a separate kernel type without further defining it.